

B. Tech Degree IV Semester Examination April 2011**ME 405 HYDRAULIC MACHINERY***(2006 Scheme)*

Time : 3 Hours

Maximum Marks : 100

PART A*(Answer ALL questions)***(8 x 5 = 40)**

- I. (a) What is meant by geometric, kinematic and dynamic similarities?
 (b) Obtain an expression for the force exerted by a jet of water on the inclined plate moving in the direction of jet.
 (c) Draw inlet and outlet velocity triangles for a pelton turbine and indicate the direction of various velocities and obtain an expression for the work done per second by water on the runner of a Pelton Wheel.
 (d) Derive an expression for specific speed in turbines.
 (e) Explain the working of a single stage centrifugal pump with neat sketches.
 (f) What is an air vessel? Describe the function of the air vessel for reciprocating pumps.
 (g) What is a hydraulic intensifier? Explain its principle and working.
 (h) What is vortex amplifier? Explain its working.

PART B**(4 x 15 = 60)**

- II. The pressure difference ' ΔP ' in a pipe of diameter ' D ' and length ' ℓ ' due to viscous flow depends on the velocity ' V ' viscosity ' μ ' and density ' δ '. Using Buckingham's π - theorem obtain an expression for ΔP . **(15)**

OR

- III. (a) Show that the efficiency of a free jet striking normally on a series of flat plates mounted on a periphery of a wheel can never exceed 50%. **(8)**
 (b) A jet of water having a velocity of 40m/s strikes a curved vane, which is moving with a velocity of 20m/s. The jet makes an angle of 30° with the direction of motion of vane at inlet and leaves at an angle of 90° to the direction of motion of vane at outlet. Draw the velocity triangles at inlet and outlet and determine the vane angles at inlet and outlet so that the water enters and leaves the vane without shock. **(7)**
- IV. (a) Differentiate between the impulse and reaction turbines. **(5)**
 (b) A Kaplan turbine develops 24647.6KW power at an average head of 39 meters. Assuming a speed ratio of 2, flow ratio of 0.6, diameter of boss equal to 0.35 times the diameter of the runner and an overall efficiency of 90%. Calculate the diameter, speed and specific speed of the turbine. **(10)**

OR

- V. (a) What are unit quantities? Obtain an expression for unit speed, unit discharge and unit power for a turbine. **(5)**
 (b) A conical draft tube having inlet and outlet diameters 1m and 1.5m, discharges water at outlet with a velocity of 2.5m/s. The total length of the draft tube is 6m and 1.2m of the length of draft tube is immersed in water. If the atmospheric pressure head is 10.3m of water and loss of head due to friction in the draft tube is equal to $0.2 \times$ velocity head at the outlet of the tube find
 (i) Pressure head at inlet
 (ii) Efficiency of draft tube **(10)**
- VI. (a) Obtain an expression for minimum speed for starting a centrifugal pump. **(5)**
 (b) A centrifugal pump having outer diameter of impeller is equal to two times the inner diameter and running at 1000 rpm works against a total head of 40m. The velocity of flow through the impeller is constant and equal to 2.5 m/s. The vanes are set back at an angle of 40° at outlet. If the outer diameter of the impeller is 500mm and width at outlet is 50mm. Determine:
 (i) Vane angle at inlet
 (ii) Work done by impeller on water per second
 (iii) Manometric efficiency **(10)**

OR**(P.T.O.)**

- VII. What is the effect of acceleration of the piston on the velocity and acceleration of the water in suction and delivery pipes of a reciprocating pump? Obtain an expression for the pressure head due to acceleration in the suction and delivery pipes. (15)
- VIII. Explain with the help of a neat sketch the principle and working of the following hydraulic devices. (15)
- (i) Hydraulic lift
 - (ii) Hydraulic coupling
 - (iii) Hydraulic torque converter
- OR**
- IX. With neat sketches explain the working of (15)
- (i) Vane pump
 - (ii) Gear pump
 - (iii) Radial flow pump